

AIGR Deep Research Report — *pvdD* (Q88F79, PP_4219), *Pseudomonas putida* KT2440

Focus: Function-assignment hypothesis — "PvdD participates in monocarboxylic acid biosynthesis (GO:0072330)"

Organism: *Pseudomonas putida* KT2440 (NCBITaxon:160488) · **Gene:** pvdD · **UniProt:** Q88F79 · **Locus:** PP_4219 · **Focus type:** function_assignment

Summary

The seed hypothesis proposes that *P. putida* KT2440 PvdD should be curated under **GO:0072330 "monocarboxylic acid biosynthetic process"** because, in building its native pyoverdine, it produces molecules that could fall within the broad monocarboxylic-acid chemical class. Importantly, the hypothesis asks the reviewer to do the honest chemistry — to count the *free* carboxyl groups of the *actual* KT2440 products (PVD G4R and G4R A, the succinamide/cyclized and succinate/linear variants) rather than inferring the count from amino-acid names — and to consider that "being a pyoverdine rather than an enterobactin synthetase does not alone exclude this broad chemical class."

Performing exactly that structural accounting **refutes** the hypothesis. The native KT2440 pyoverdine is a **polycarboxylic** chromopeptide siderophore carrying **2–3 free carboxylic acid groups** (Asp β -COOH, threo- β -hydroxy-Asp β -COOH, and — in the succinate variant only — the free succinate carboxyl). A molecule with two or three free carboxyls is, by the term's own definition, **not** a monocarboxylic acid. The chemical-class defense the hypothesis invites therefore fails on chemistry, not on taxonomy. Separately, PvdD's *direct* molecular activity is nonribosomal peptide synthesis — it is the terminal, chain-releasing NRPS subunit — and it does not itself synthesize the succinyl side chain (the fragment that most resembles a small monocarboxylic acid); that acyl group is installed by separate maturation enzymes.

The annotation itself is also weak and unstable. **All nine current GO annotations on Q88F79 are IEA** (electronic, zero experimental). GO:0072330 specifically originates from **UniProtKB-ARBA** association-rule automation and is **absent from the live GOA/QuickGO extract**, where the corresponding broad automated process term is instead GO:0044550 "secondary metabolite biosynthetic process." GO:0072330 co-occurs with a cluster of clearly erroneous **enterobactin**-branch TreeGrafter mismaps — a textbook signature of NRPS-family paralog over-annotation. The correct, biologically supported terms are **GO:0002049 "pyoverdine biosynthetic process"** and **GO:0019290 "siderophore biosynthetic process"** (BP), with **GO:0031177 "phosphopantetheine binding"** retained (MF). **Verdict: REFUTED / OVER-ANNOTATED.**

Key Findings

Finding 1 — The native KT2440 pyoverdine is a polycarboxylic chromopeptide siderophore, not a monocarboxylic acid

The hypothesis is right to demand a genuine free-carboxyl count from the *actual* structures. When that accounting is performed, it defeats the hypothesis rather than supporting it.

Structural mass-spectrometry studies define the real KT2440 products — the pyoverdines **PVD G4R** and **G4R A** — as a three-part architecture: (i) a dihydroxyquinoline-type chromophore, (ii) a heptapeptide chain **Asp–Orn–OHAsp–Dab–Gly–Ser–cOHOrn**, and (iii) a C-4 diacid side chain attached to the chromophore (succinic acid or its monoamide/"succinamide" derivative; a malamide/malic variant is reported for the related *P. taiwanensis* strain, not KT2440). [PMID: 25895945](#) states that the pyoverdine "consists of a dihydroxyquinoline-type chromophore, a peptide chain of variable length and conformation, and a **side chain composed of a dicarboxylic acid or its monoamide derivative**," and that "the three PVDs of *P. putida* possess the same peptide chain of seven amino acids, **Asp–Orn–OHAsp–Dab–Gly–Ser–cOHOrn**, with a cyclicized portion present in two of the PVDs."

Free-carboxyl accounting:

Carboxyl source	Chemical status	Free -COOH?
Asp β-carboxyl	free carboxylic acid	☐ 1
OHAsp β-carboxyl	free carboxylic acid (also α-hydroxy-acid Fe ligand)	☐ 1
Side-chain dicarboxylic acid (succinate variant)	free carboxylic acid	☐ 1 (0 in succinamide/monoamide variant)
Chromophore C1-carboxyl	amide -bonded to the peptide	☐ (amide, not acid)
C-terminal cyclo-hydroxyornithine (cOHOrn)	cyclic hydroxamate	☐ (not a carboxylic acid)

Thus the molecule carries **2–3 free carboxylic acid groups** — it is **polycarboxylic**. Its iron-binding triad is **catecholate (chromophore) + hydroxamate (cOHOrn) + α-hydroxycarboxylate (OHAsp)**, a classic mixed-ligand siderophore rather than a simple organic acid. Product identity is independently confirmed by [PMID: 28631237](#): "In *Pseudomonas* sp. *putida* KT2[2]40 samples, the known pyoverdines **G4R and G4R A** were readily confirmed."

Because GO:0072330 is defined by a product class carrying **exactly one** carboxyl group, a molecule with two or three free carboxyls is excluded by the term's own definition. The hypothesis's chemical-class reasoning, applied rigorously, lands on refutation.

Finding 2 — GO:0072330 on Q88F79 is automated ARBA over-annotation; the direct function is the terminal NRPS module of pyoverdine biosynthesis

The UniProt Q88F79 entry describes a **3,470-amino-acid** multi-modular protein with Pfam architecture **Condensation (PF00668) + AMP-binding/adenylation (PF00501) + PP-binding/thiolation-PCP (PF00550) + a second AMP-binding module + Thioesterase (PF00975)** — the canonical signature of a **terminal, chain-releasing NRPS subunit** that adds the final residues and releases/cyclizes the mature peptide.

The direct functional evidence comes from the well-characterized *P. aeruginosa* ortholog. [PMID: 25015884](#) establishes that "the *Pseudomonas aeruginosa* nonribosomal peptide synthetase (NRPS) **PvdD contains two modules that each incorporate an l-threonine residue at the C-terminal end of pyoverdine.**" [PMID: 26268580](#) reinforces the mechanism: "**Non-ribosomal peptide synthetase (NRPS) enzymes govern the assembly of amino acids and related monomers into peptide-like natural products.**" PvdD's immediate biochemical acts are therefore ATP-dependent amino-acid adenylation, phosphopantetheine-mediated

thiolation, condensation (peptide-bond formation), and thioesterase-catalyzed release/macrocyclization — **nonribosomal peptide synthesis**, not the biosynthesis of a free monocarboxylic acid.

Within the IEA-only GO panel, GO:0072330 is an **IEA:UniProtKB-ARBA** call. The remaining machine annotations include an **enterobactin cluster that is simply wrong** for a pyoverdine synthetase — GO:0009366 "enterobactin synthetase complex," GO:0047527 "2,3-dihydroxybenzoate–serine ligase activity," and GO:0009239 "enterobactin biosynthetic process" — all TreeGrafter mismaps arising because both enterobactin and pyoverdine systems are NRPS-family. Only **GO:0031177 (phosphopantetheine binding)** and the generic amino-acid-activation concept are appropriate; GO:0072330 is not.

Finding 3 — GO definitions and live QuickGO provenance confirm GO:0072330 is inapt automated annotation; all Q88F79 evidence is IEA

To make the decision auditable, term definitions and the live annotation set were pulled directly from EBI QuickGO.

- **GO:0072330 "monocarboxylic acid biosynthetic process"** = "the chemical reactions and pathways resulting in the **formation of monocarboxylic acids, any organic acid containing one carboxyl (–COOH) group.**" The KT2440 pyoverdine has 2–3 free carboxyls → **fails** the single-COOH definition. Reinforced by [PMID: 25895945](#): the side chain is "**a dicarboxylic acid or its monoamide derivative.**"
- **GO:0002049 "pyoverdine biosynthetic process"** = "formation of the siderochrome pyoverdine" → **exact fit.**
- **GO:0019290 "siderophore biosynthetic process"** → fits as the parent process.
- **GO:0043041** ("amino acid activation for nonribosomal peptide biosynthetic process") is now **OBSOLETE** — cannot be retained verbatim.

A live QuickGO extract for Q88F79 returned 9 annotations, ALL IEA (ECO:0000501 UniProt or ECO:0007826 TreeGrafter), **zero experimental**:

Aspect	GO ID	Term	Evidence / source	Assessment
MF	GO:0003824	catalytic activity	IEA (UniProt)	Correct but generic
MF	GO:0031177	phosphopantetheine binding	IEA (UniProt)	Correct — retain
MF	GO:0047527	2,3-DHB–serine ligase activity	IEA (TreeGrafter)	Wrong (enterobactin mismap)
BP	GO:0009239	enterobactin biosynthetic process	IEA (TreeGrafter)	Wrong (enterobactin mismap)
BP	GO:0043041	amino acid activation for NRP biosynthesis	IEA	Obsolete
BP	GO:0044550	secondary metabolite biosynthetic process	IEA (UniProt)	Correct but broad
CC	GO:0005737	cytoplasm	IEA	Plausible
CC	GO:0005829	cytosol	IEA	Plausible
CC	GO:0009366	enterobactin synthetase complex	IEA (TreeGrafter)	Wrong (enterobactin mismap)

Notably, **GO:0072330** appears in the UniProtKB entry (ARBA prediction) but is NOT in the current GOA/QuickGO extract, where the broad automated BP is instead GO:0044550. This instability — the term flickering between resources across releases — is itself diagnostic of a low-confidence, non-curated automated call that should not be promoted into a curated review.

Mechanistic Model / Interpretation

The refutation rests on separating three levels: the **enzyme's direct reaction**, the **product's chemical class**, and the **automated annotation provenance**. All three point away from GO:0072330.

DIRECT MOLECULAR FUNCTION (what PvdD actually catalyzes)

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PvdD = terminal NRPS subunit (3,470 aa)
[C] – [A] – [T/PCP] – [A] – [T/PCP] – [TE]
condensation · adenylation · thiolation · thioesterase
→ activates aa, forms peptide bonds, releases C-terminus
MF: G0:0031177 (Ppant binding); NRPS / amino-acid activation

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| assembles



PRODUCT (native KT2440 pyoverdine: PVD G4R / G4R A)

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Chromophore (dihydroxyquinoline, catecholate Fe ligand)
| C1-carboxyl = AMIDE to peptide (not free acid)
| side chain = succinate / succinamide / malamide
Peptide: Asp–Orn–OHAsp–Dab–Gly–Ser–cOHOrn
Asp β-COOH ..... FREE acid (1)
OHAsp β-COOH ..... FREE acid + α-OH-acid Fe ligand (1)
cOHOrn ..... CYCLIC HYDROXAMATE (not acid)
Side-chain COOH (succinate variant only) .... FREE acid (1)
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Σ FREE carboxyls = 2–3 → POLYCARBOXYLIC

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| therefore



CHEMICAL-CLASS TEST against G0:0072330

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G0:0072330 = "one carboxyl (-COOH) group" (monocarboxylic)
Product carries 2–3 free carboxyls → DOES NOT MATCH
Correct BP: G0:0002049 pyoverdine biosynthetic process
             G0:0019290 siderophore biosynthetic process

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The key logical trap the seed sets — "do not infer the count solely from amino-acid names" and "the broad chemical class is not excluded just because it's a pyoverdine" — is answered by doing the accounting properly. The molecule is polycarboxylic, so it fails the **monocarboxylic** definition. The broad-class defense would apply only if the product had exactly one free carboxyl; it does not. And even under the most permissive reading, PvdD does not *synthesize* the succinyl acyl group (installed by separate acylation/tailoring machinery), so attributing free-acid production to PvdD's catalytic activity is doubly unsupported. PvdD's thioesterase-mediated macrocyclization (to cyclo-OHOrn) in fact *removes* a free carboxyl by lactamization rather than creating a monocarboxylic acid.

Evidence Base (Evidence Matrix)

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence limitation
PMID: 25895945	Direct structural (LC– HR-MS/MS)	Refutes	Product carboxyl count	Pyoverdine = dihydroxyquinoline chromophore + Asp-Orn-OHAsp- Dab-Gly-Ser- cOHOrn peptide + dicarboxylic-acid or monoamide side chain → polycarboxylic	<i>P. putida</i> KT2440 supernatant	High for structure; carboxyl is interpretat of reporte moieties; count var by varian but alway
PMID: 28631237	Direct structural (LC– HR-MS/MS)	Supports (product identity)	Which pyoverdines does KT2440 make?	Confirms G4R and G4R A as native KT2440 pyoverdines	<i>P. putida</i> KT2440 & <i>P. taiwanensis</i> VLB120	High; the malamide OHAsn variant reported i for <i>P. taiwanens</i> not KT244
PMID: 25015884	Mutant/ biochemical (domain swaps; ortholog)	Supports (true function) / Refutes scope	What does PvdD catalyze?	<i>P. aeruginosa</i> PvdD = two-module terminal NRPS adding C-terminal residues of pyoverdine	<i>P. aeruginosa</i> ; ortholog of Q88F79	High for mechanis transfer b orthology (KT2440 C terminus Ser/cOHOrn strain- specific)
PMID: 26268580	Biochemical / domain- engineering	Supports (function class)	Is PvdD an NRPS?	NRPS enzymes assemble amino acids into peptide- like products ; thiolation-domain portability in pyoverdine NRPS	Pyoverdine NRPS systems	High for NRPS classificat

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence limitation
PMID: 19459056	Structural / physiological	Qualifies (product identity)	Does KT2440 make a single characterized pyoverdine?	KT2440 produces "an already characterised pyoverdine"; no second siderophore	<i>P. putida</i> KT2440 vs <i>P.</i> <i>entomophila</i>	Medium-L orients on siderophore biology
UniProt Q88F79 / EBI QuickGO (this run)	Computational / database (definition + provenance)	Refutes	Is GO:0072330 apt, supported, stable?	GO:0072330 = single-COOH class (product fails it); Q88F79 has 9/9 IEA annotations, 0 experimental; GO:0072330 in UniProt but absent from live GOA (GO:0044550 there instead); enterobactin terms are mismaps	GOA/ QuickGO, 2026	High for provenance snapshot time

How the literature supports or challenges the findings

- **Structural characterization of multiple pyoverdines...** [PMID: 25895945](#) — Primary structural anchor defining the KT2440 scaffold and the dicarboxylic/monoamide side chain. **Challenges** the monocarboxylic classification by demonstrating multiple carboxyl groups.
- **Structural characterization of pyoverdines produced by *P. putida* KT2440...** [PMID: 28631237](#) — Independently confirms G4R/G4R A identity. **Supports** the product identity used in the accounting.
- **Biosynthesis of novel Pyoverdines by domain substitution in an NRPS of *P. aeruginosa*.** [PMID: 25015884](#) — Direct functional characterization of PvdD as a two-module terminal NRPS in the ortholog. **Supports** the true molecular function.
- **Portability of the thiolation domain in recombinant pyoverdine NRPS.** [PMID: 26268580](#) — Establishes NRPS mechanism/engineering for pyoverdine synthetases. **Supports** the NRPS classification.

- **Siderophore-mediated iron acquisition in *P. entomophila* L48 and *P. putida* KT2440.**
[PMID: 19459056](#) — Physiological context; KT2440 makes a single characterized pyoverdine and no second siderophore. **Orients** the siderophore biology.

GO Curation Implications (leads — require curator verification)

- **GO:0072330 (monocarboxylic acid biosynthetic process, BP):** *Do not curate / recommend removal of reliance on it.* IEA-ARBA only, chemically inapt (product is polycarboxylic), unstable across releases, and not PvdD's direct activity. Not suitable as a positive curation statement.
- **GO:0009366 / GO:0047527 / GO:0009239 (enterobactin complex / DHB-Ser ligase / enterobactin biosynthesis):** *Remove* — wrong pathway/product (pyoverdine ≠ enterobactin); NRPS-family paralog mismap.
- **GO:0044550 (secondary metabolite biosynthetic process, BP):** present in live GOA (IEA:UniProt); acceptable as a broad parent but should be superseded by the specific siderophore/pyoverdine terms.
- **GO:0043041 (BP):** now **OBSOLETE** — do not retain verbatim; replace with a current NRPS/pyoverdine term.
- **Retain / prefer instead:**
- **MF: GO:0031177 phosphopantetheine binding** (retain); consider adenylation/ligase and NRPS thioesterase activity terms supported by domain content (GO:0003824 "catalytic activity" is too generic).
- **BP:** add the specific **GO:0002049 pyoverdine biosynthetic process** and/or **GO:0019290 siderophore biosynthetic process**.
- **CC:** cytoplasm/cytosol (GO:0005737/GO:0005829) plausible; remove enterobactin synthetase complex (GO:0009366).
- Avoid "protein binding" as an endpoint; specific NRPS/siderophore terms are supported.

Mechanistic Scope

Direct activity (what PvdD does): ATP-dependent amino-acid adenylation → phosphopantetheine-mediated thiolation → condensation (peptide-bond formation) → thioesterase-catalyzed release/macrocyclization of the C-terminal pyoverdine peptide. This is captured by NRPS/peptide-biosynthesis MF concepts and by **GO:0002049** / **GO:0019290** at the process level; **GO:0031177** is directly supported by the PP-binding/PCP domains.

Out of scope (downstream / other enzymes): chromophore formation, N-acylation with the succinate/malate side chain, side-chain amidation, and iron chelation. The **free carboxyls** in the finished siderophore are properties of the *assembled multi-enzyme product* and of separate building-block and tailoring enzymes — **not** a monocarboxylic-acid product of PvdD. Iron acquisition and growth/competition phenotypes are downstream physiological consequences, not PvdD's molecular function.

Conflicts and Alternatives

1. **Paralog confusion:** the enterobactin annotations and the ARBA monocarboxylic-acid rule both derive from the generic NRPS/AMP-binding domain, not from PvdD-specific evidence.
2. **Strain-specificity:** the malic-acid side chain and the Orn-Asp-OHAsn-Thr... sequence emphasized by [PMID: 28631237](#) pertain to *P. taiwanensis* VLB120; the KT2440 backbone is Asp-Orn-OHAsp-Dab-Gly-Ser-cOHOrn ([PMID: 25895945](#)). Do not conflate.
3. **Product-variant ambiguity:** succinate vs. succinamide (monoamide) vs. malamide variants shift the exact free-carboxyl count (2 vs. 3), but every variant is polycarboxylic (≥ 2 free – COOH), so the monocarboxylic classification fails regardless.
4. **Charitable reading:** even if the succinate side chain is formally a monocarboxylic acyl group, that acyl group is not synthesized by PvdD, so the term still fails for PvdD specifically.
5. **The seed's "subsequent" reference:**

P 31451546

(arginine biosynthesis modulating pyoverdine production under oxidative-stress adaptation) is a physiology/genetics paper — it provides **no** pyoverdine structural/carboxyl data and does **not** support a monocarboxylic-acid function for PvdD.

6. **Annotation instability:** GO:0072330 (UniProtKB-ARBA) vs GO:0044550 (live GOA) for the same protein shows the broad automated BP is version-dependent and should not drive curation.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
No experimental GO evidence for Q88F79 itself	Live QuickGO extract: 9/9 IEA, 0 experimental	The entire GO panel is automation; any curated term needs a defensible code	Direct KT2440 PvdD assay or curated ISS/IBA transfer from <i>P. aeruginosa</i> PvdD
Exact free-carboxyl count under physiological/iron-bound state	Structural definitions from  / 28631237	Chelation may deprotonate/engage the OHAsp carboxyl; affects count (2 vs 3), not the mono-vs-poly conclusion	pH-dependent titration / crystal or NMR structure of Fe-pyoverdine
Which enzyme installs the succinyl/free acid	Domain architecture of PvdD (no acyltransferase)	Confirms PvdD does not make the free-acid moiety, weakening any monocarboxylic claim	Pathway mapping of the pyoverdine acylation/tailoring step (PvdN/PvdO/PvdP-type enzymes)
KT2440 PvdD precise module/substrate assignment	Ortholog (<i>P. aeruginosa</i>) + domain architecture	Determines which residues PvdD adds in KT2440	KT2440-specific NRPS module/substrate mapping or ΔpvdD product analysis
ARBA rule provenance for GO:0072330	Observed term in UniProt, absent in GOA	Understanding why ARBA fired helps prevent re-annotation	Inspect the specific ARBA association rule and its features

Discriminating Tests

1. **Curator provenance check (fastest, decisive):** Confirm in QuickGO/GOA that GO:0072330 is IEA:UniProtKB-ARBA and absent from the curated GOA extract for Q88F79. This alone justifies not promoting it.
 2. **Structural free-COOH audit:** For each KT2440 variant (G4R, G4R A, succinamide, malamide), enumerate free carboxyls from published structures; enumerate protonation-state-resolved carboxyls of apo- vs ferri-pyoverdine. Expected: ≥ 2 in all cases → fails monocarboxylic definition.
 3. **Domain-architecture verification:** Re-run InterPro/Pfam on Q88F79 to confirm C-A-T-A-T-TE terminal-NRPS architecture and PP-binding domains (supporting GO:0031177), and the absence of enterobactin-specific domains that would justify the mismapped terms.
 4. **Ortholog cross-check:** Align Q88F79 with *P. aeruginosa* PvdD to confirm orthology and transfer the two-module terminal-NRPS function ([PMID: 25015884](#)) as ISS/IBA.
 5. **Mutant experiment (definitive in-organism):** Delete *pvdD* in KT2440 and confirm loss/truncation of PVD G4R/G4R A by LC-HR-MS/MS, establishing GO:0002049 with experimental evidence, plus complementation to show PvdD builds the C-terminal peptide (not the acyl side chain).
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Proposed Follow-up Actions / Curation Leads (verify before applying)

- **Action:** Reject the hypothesis as a basis to assert GO:0072330; flag GO:0072330 and all enterobactin terms as IEA over-annotations.
- **Candidate references (with snippets to verify):**
 - [PMID: 25895945](#): "*side chain composed of a dicarboxylic acid or its monoamide derivative*"; "*the same peptide chain of seven amino acids, **Asp-Orn-OHAsp-Dab-Gly-Ser-cOHOrn.*" → polycarboxylic structure.
 - [PMID: 28631237](#): "*the known pyoverdines G4R and G4R A were readily confirmed.*" → product identity.
 - [PMID: 25015884](#): "*PvdD contains two modules that each incorporate an l-threonine residue at the C-terminal end of pyoverdine.*" → true molecular function.

- **PMID: 26268580:** "NRPS enzymes govern the assembly of amino acids... into peptide-like natural products." → NRPS classification.
- **Candidate replacement / new GO terms:** GO:0002049 (pyoverdine biosynthetic process), GO:0019290 (siderophore biosynthetic process); retain GO:0031177 (phosphopantetheine binding). Note GO:0043041 is obsolete.
- **Suggested curator questions:** Does any source provide experimental (EXP/IDA/IMP) support for GO:0072330 on PvdD? Should broad chemical-class BP terms be suppressed when a specific pathway term exists?
- **Suggested experiments:** Δ pvdD deletion + LC–HR-MS/MS pyoverdine readout for in-organism experimental evidence; InterPro/Pfam re-scan to document domain support for GO:0031177 and absence of enterobactin domains.

Conclusion

The hypothesis that PvdD (Q88F79) should be curated under **GO:0072330 "monocarboxylic acid biosynthetic process"** is **refuted**. The native KT2440 pyoverdine is a **polycarboxylic** chromopeptide siderophore (2–3 free –COOH), which fails the term's single-carboxyl definition; the annotation is unsupported, unstable **IEA:UniProtKB-ARBA** automation co-occurring with erroneous enterobactin mismaps; and PvdD's direct function is **nonribosomal peptide synthesis** (terminal NRPS module of the pyoverdine synthetase). The curation should instead use **GO:0002049 (pyoverdine biosynthetic process)** and **GO:0019290 (siderophore biosynthetic process)**, retain **GO:0031177 (phosphopantetheine binding)**, and remove the erroneous enterobactin terms (GO:0009366, GO:0047527, GO:0009239).